



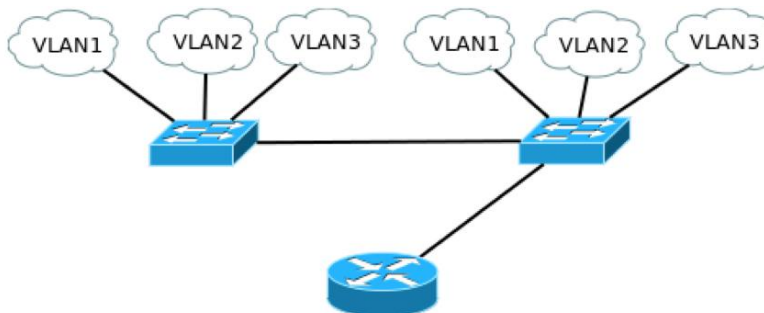
Guia 3– Inter-VLAN Routing

Objectivos:

- Definição de VLANs
- Routing entre VLANs

Inter-VLAN Routing: Router-on-a-Stick

3. Monte a rede apresentada na figura, onde foi adicionado um router.



- Configure o router para suportar sub-interfaces e routing Inter-VLAN (802.1Q):

```
Router(config)# interface FastEthernet0/0
```

```
Router(config-if)# no shutdown
```

```
Router(config-if)# interface FastEthernet0/0.1
```

```
Router(config-if)# encapsulation dot1Q 1 native      !VLAN1
```

```
Router(config-if)# ip address 10.1.1.1 255.255.255.0
```

```
!
```

```
Router(config-if)# interface FastEthernet0/0.2
```

```
Router(config-if)# encapsulation dot1Q 2          !VLAN2
```

```
Router(config-if)# ip address 10.2.2.1 255.255.255.0
```

```
!
```

```
Router(config-if)# interface FastEthernet0/0.3
```

```
Router(config-if)# encapsulation dot1Q 3          !VLAN3
```

```
Router(config-if)# ip address 10.3.3.1 255.255.255.0
```

Verifique a tabela de encaminhamento. Coloque terminais nas diferentes VLANs, configure neles a respetiva gateway (ou seja, os endereços IP das sub-interfaces do router) e teste a conectividade. Capture os pacotes trocados entre o router e o switch (da direita).

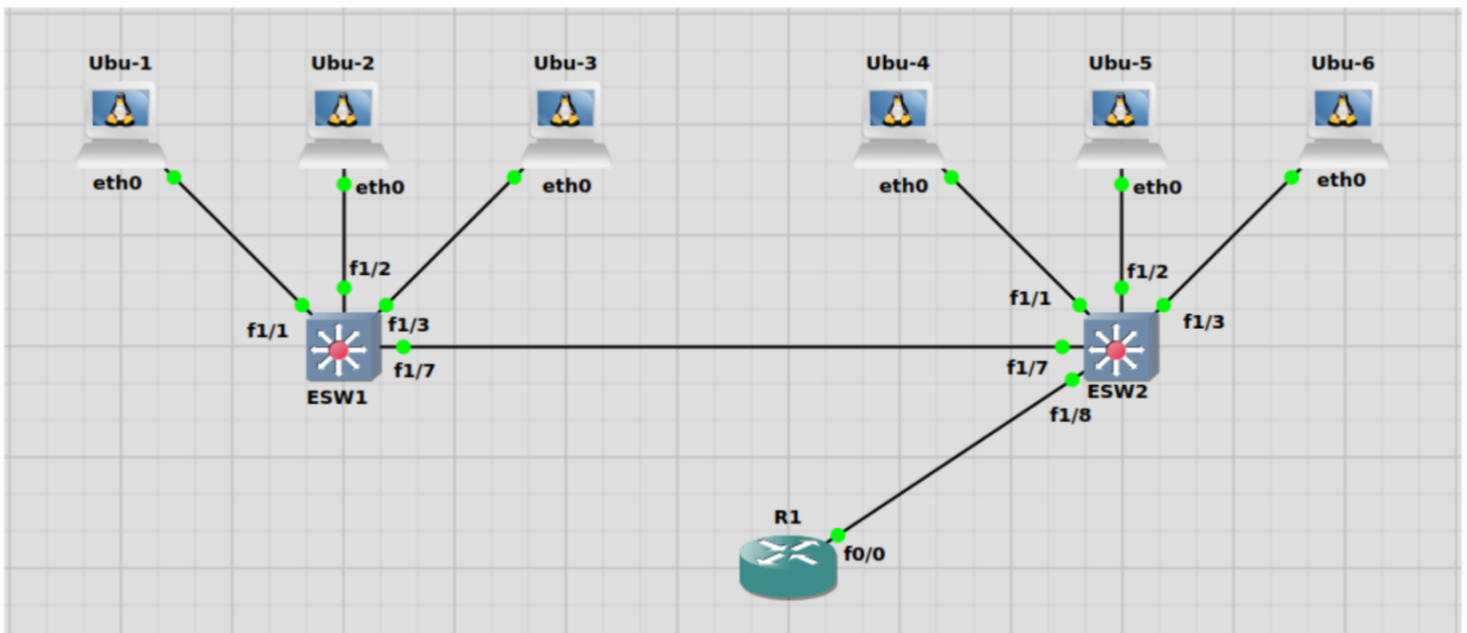
- **Dica1:** Vai precisar de ativar mais um trunk algures.
- **Dica2:** Se utilizou routers para substituir os hosts, não se esqueça de incluir a rota por defeito para fora da rede: `ip route 0.0.0.0 0.0.0.0 endereço_da_gateway`
- **Dica3:** Se tiver que reiniciar o GNS3 não se esqueça que tem que reativar a base de dados das vlans.

Cofinanciado por:





5. Grave as configurações dos equipamentos, bem como o projecto.



Ubu-1:

```
root@Ubu-1:~# cat /etc/network/interfaces
```

```
# Static config for eth0
auto eth0
iface eth0 inet static
    address 10.1.1.2
    netmask 255.255.255.0
    gateway 10.1.1.1
```

Ubu-4:

```
root@Ubu-4:~# cat /etc/network/interfaces
```

```
# Static config for eth0
auto eth0
iface eth0 inet static
    address 10.1.1.3
    netmask 255.255.255.0
    gateway 10.1.1.1
```

Ubu-2:

```
root@Ubu-2:~# cat /etc/network/interfaces
```

```
# Static config for eth0
auto eth0
iface eth0 inet static
    address 10.2.2.2
    netmask 255.255.255.0
    gateway 10.2.2.1
```

Ubu-5:

```
root@Ubu-5:~# cat /etc/network/interfaces
```

```
# Static config for eth0
auto eth0
iface eth0 inet static
    address 10.2.2.3
    netmask 255.255.255.0
    gateway 10.2.2.1
```

Ubu-3:

```
root@Ubu-3:~# cat /etc/network/interfaces
```

```
# Static config for eth0
auto eth0
iface eth0 inet static
    address 10.3.3.2
    netmask 255.255.255.0
    gateway 10.3.3.1
```


Ubu-6:

```
root@Ubu-6:~# cat /etc/network/interfaces
```

```
# Static config for eth0
auto eth0
iface eth0 inet static
    address 10.3.3.3
    netmask 255.255.255.0
    gateway 10.3.3.1
```

```
ESW1#vlan database
ESW1(vlan)#vlan 1
VLAN 1 modified:
ESW1(vlan)#vlan 2
VLAN 2 added:
    Name: VLAN0002
ESW1(vlan)#vlan 3
VLAN 3 added:
    Name: VLAN0003
ESW1(vlan)#exit
APPLY completed.
Exiting....
```

```
ESW1#conf t
Enter configuration commands, one per line. End with
ESW1(config)#int f1/1
ESW1(config-if)#switchport
ESW1(config-if)#switchport mode access
ESW1(config-if)#switchport mode access
ESW1(config-if)#switchpo
ESW1(config-if)#switchport access
ESW1(config-if)#switchport access vlan 1
ESW1(config-if)#int f1/2
ESW1(config-if)#switchport mode access
ESW1(config-if)#switchport access vlan 2
ESW1(config-if)#int f1/3
ESW1(config-if)#switchport mode access
ESW1(config-if)#switchport access vlan 3
ESW1(config-if)#int f1/7
ESW1(config-if)#shutdown
    ^
% Invalid input detected at '^' marker.

ESW1(config-if)#shutdow
ESW1(config-if)#shutdown
ESW1(config-if)#
*Mar  1 00:09:55.619: %LINK-5-CHANGED: Interface FastEthernet0/7 is now
ratively down
*Mar  1 00:09:56.619: %LINEPROTO-5-UPDOWN: Line protocol is now
ged state to down
ESW1(config-if)#switchport mode trun
ESW1(config-if)#switchport mode trunk
ESW1(config-if)#switchport trunk enca
ESW1(config-if)#switchport trunk encapsulation dot
ESW1(config-if)#switchport trunk encapsulation dot1q
ESW1(config-if)#no shut
ESW1(config-if)#no shutdown
ESW1(config-if)#
```

```
ESW1#sh vlan-switch
```

VLAN	Name	Status	Ports
1	default	active	Fa1/0, Fa1/1, Fa1/4, Fa1/5 Fa1/6, Fa1/8, Fa1/9, Fa1/10 Fa1/11, Fa1/12, Fa1/13, Fa1/14 Fa1/15
2	VLAN0002	active	Fa1/2
3	VLAN0003	active	Fa1/3
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	1002	1003
2	enet	100002	1500	-	-	-	-	-	0	0
3	enet	100003	1500	-	-	-	-	-	0	0
1002	fddi	101002	1500	-	-	-	-	-	1	1003
1003	tr	101003	1500	1005	0	-	-	srb	1	1002
1004	fdnet	101004	1500	-	-	1	-	ibm	0	0
1005	trnet	101005	1500	-	-	1	-	ibm	0	0

```
ESW2#conf t
```

```
Enter configuration commands, one per line. End with CNTL/Z.
```

```
ESW2(config)#int f1/1
```

```
ESW2(config-if)#switc
```

```
ESW2(config-if)#switchport mode access
```

```
ESW2(config-if)#switchport mode access
```

```
ESW2(config-if)#switchport acce
```

```
ESW2(config-if)#switchport access vlan 1
```

```
ESW2(config-if)#int f1/2
```

```
ESW2(config-if)#switchport mode access
```

```
ESW2(config-if)#switchport access vlan 2
```

```
%Access VLAN 2 does not exist. Please add it to vlan database
```

```
ESW2(config-if)#int f1/3
```

```
ESW2(config-if)#switchport mode access
```

```
ESW2(config-if)#switchport access vlan 3
```

```
%Access VLAN 3 does not exist. Please add it to vlan database
```

```
ESW2(config-if)#int f1/7
```

```
ESW2(config-if)#shut
```

```
ESW2(config-if)#shutdown
```

```
ESW2(config-if)#
```

```
*Mar 1 00:03:06.843: %LINK-5-CHANGED: Interface FastEthernet1/7, changed state to administrati  
vely down
```

```
*Mar 1 00:03:07.843: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/7, changed  
state to down
```

```
ESW2(config-if)#switchport mode trunk
```

```
ESW2(config-if)#switchport trunk
```

```
ESW2(config-if)#switchport trunk enc
```

```
ESW2(config-if)#switchport trunk encapsulation dot
```

```
ESW2(config-if)#switchport trunk encapsulation dot1q
```

```
ESW2(config-if)#no shut
```

```
ESW2(config-if)#
```

```
*Mar 1 00:03:26.639: %DTP-5-TRUNKPORTON: Port Fa1/7 has become dot1q trunk
```

```
ESW2(config-if)#switchport trunk encapsulation dot1q
```

```
*Mar 1 00:03:29.135: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/7, changed  
state to up
```

```
ESW2(config-if)#int f1/8
```

```
ESW2(config-if)#switchport mode trunk
```

```
ESW2(config-if)#switchport trunk encapsulation dot1q
```

```
ESW2(config-if)#shut
```

```
ESW2(config-if)#n
```

```
*Mar 1 00:06:24.627: %LINK-5-CHANGED: Interface FastEthernet1/8, changed state to administrati  
vely down
```

```
ESW2(config-if)#no shut
```

```
ESW2#sh vlan-switch
```

VLAN	Name	Status	Ports
1	default	active	Fa1/0, Fa1/1, Fa1/4, Fa1/5 Fa1/6, Fa1/9, Fa1/10, Fa1/11 Fa1/12, Fa1/13, Fa1/14, Fa1/15
2	VLAN0002	active	Fa1/2
3	VLAN0003	active	Fa1/3
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	1002	1003
2	enet	100002	1500	-	-	-	-	-	0	0
3	enet	100003	1500	-	-	-	-	-	0	0
1002	fddi	101002	1500	-	-	-	-	-	1	1003
1003	tr	101003	1500	1005	0	-	-	srb	1	1002
1004	fdnet	101004	1500	-	-	1	-	ibm	0	0
1005	trnet	101005	1500	-	-	1	-	ibm	0	0

```
ESW2#sh interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Fa1/7	on	802.1q	trunking	1
Fa1/8	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Fa1/7	1-1005
Fa1/8	1-1005

Port	Vlans allowed and active in management domain
Fa1/7	1-3
Fa1/8	1-3

Port	Vlans in spanning tree forwarding state and not pruned
Fa1/7	1
Fa1/8	1

3.

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int f0/0
R1(config-if)#no shut
R1(config-if)#int
*Mar  1 00:01:32.851: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar  1 00:01:33.851: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1(config-if)#int f0/0.1
R1(config-subif)#enca
R1(config-subif)#encapsulation dot
R1(config-subif)#encapsulation dot1Q 1 native
R1(config-subif)#ip address
*Mar  1 00:02:05.131: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
R1(config-subif)#ip address 10.1.1.1 255.255.255.0
R1(config-subif)#int f0/0.2
R1(config-subif)#encapsulation dot1Q 2
R1(config-subif)#ip address 10.2.2.1 255.255.255.0
R1(config-subif)#int f0/0.3
R1(config-subif)#encapsulation dot1Q 3
R1(config-subif)#ip address 10.3.3.1 255.255.255.0
```

```
R1#sh ip route
```

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

```
      10.0.0.0/24 is subnetted, 3 subnets
C      10.3.3.0 is directly connected, FastEthernet0/0.3
C      10.2.2.0 is directly connected, FastEthernet0/0.2
C      10.1.1.0 is directly connected, FastEthernet0/0.1
```

```
Ubu-1 console is now available... Press RETURN to get started.
root@Ubu-1:~# ping 10.1.1.3
PING 10.1.1.3 (10.1.1.3) 56(84) bytes of data.
64 bytes from 10.1.1.3: icmp_seq=1 ttl=64 time=4.79 ms
64 bytes from 10.1.1.3: icmp_seq=2 ttl=64 time=0.565 ms
64 bytes from 10.1.1.3: icmp_seq=3 ttl=64 time=0.606 ms
^C
--- 10.1.1.3 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2036ms
rtt min/avg/max/mdev = 0.565/1.986/4.787/1.980 ms
root@Ubu-1:~# ping 10.2.2.3
PING 10.2.2.3 (10.2.2.3) 56(84) bytes of data.
64 bytes from 10.2.2.3: icmp_seq=2 ttl=63 time=18.1 ms
64 bytes from 10.2.2.3: icmp_seq=3 ttl=63 time=21.0 ms
64 bytes from 10.2.2.3: icmp_seq=4 ttl=63 time=21.3 ms
64 bytes from 10.2.2.3: icmp_seq=5 ttl=63 time=21.3 ms
^C
--- 10.2.2.3 ping statistics ---
5 packets transmitted, 4 received, 20% packet loss, time 4030ms
rtt min/avg/max/mdev = 18.126/20.441/21.345/1.345 ms
root@Ubu-1:~# ping 10.3.3.3
PING 10.3.3.3 (10.3.3.3) 56(84) bytes of data.
64 bytes from 10.3.3.3: icmp_seq=2 ttl=63 time=11.3 ms
64 bytes from 10.3.3.3: icmp_seq=3 ttl=63 time=16.3 ms
64 bytes from 10.3.3.3: icmp_seq=4 ttl=63 time=20.7 ms
^C
--- 10.3.3.3 ping statistics ---
4 packets transmitted, 3 received, 25% packet loss, time 3016ms
rtt min/avg/max/mdev = 11.324/16.116/20.695/3.828 ms
```

Pela imagem anterior, é possível verificar que a máquina Ubu-1 tem conectividade com a Ubu-4 (e ambas estas máquinas encontram-se na VLAN 1), com a Ubu-5 (que se encontra na VLAN 2) e com a Ubu-6 (que se encontra na VLAN 3). Assim, verifica-se que existe conectividade entre as três VLANs.

Configuração do Router 1:

```
R1#sh run
```

```
Building configuration...
```

```
Current configuration : 1297 bytes
```

```
!
```

```
version 12.4
```

```
service timestamps debug datetime msec
```

```
service timestamps log datetime msec
```

```
no service password-encryption
```

```
!
```

```
hostname R1
```

```
!
```

```
boot-start-marker
```

```
boot-end-marker
```

```
!
```

```
!
```

```
no aaa new-model
```

[illegible]

```
!  
interface FastEthernet0/0.3  
  encapsulation dot1Q 3  
  ip address 10.3.3.1 255.255.255.0
```

```
!  
interface FastEthernet0/1  
  no ip address  
  shutdown  
  duplex auto  
  speed auto
```

```
!  
interface FastEthernet1/0  
  no ip address  
  shutdown  
  duplex auto  
  speed auto
```

```
!  
interface FastEthernet2/0  
  no ip address  
  shutdown  
  duplex auto  
  speed auto
```

```
!  
ip forward-protocol nd
```

```
!  
!  
no ip http server  
no ip http secure-server
```

```
!  
no cdp log mismatch duplex
```

```
!  
!  
!  
!  
control-plane
```

```
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!
```

```
line con 0
exec-timeout 0 0
privilege level 15
logging synchronous
line aux 0
exec-timeout 0 0
privilege level 15
logging synchronous
line vty 0 4
login
!
!
end
```

```
R1#sh ip int br
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	unassigned	YES	unset	up	up
FastEthernet0/0.1	10.1.1.1	YES	manual	up	up
FastEthernet0/0.2	10.2.2.1	YES	manual	up	up
FastEthernet0/0.3	10.3.3.1	YES	manual	up	up
FastEthernet0/1	unassigned	YES	unset	administratively down	down
FastEthernet1/0	unassigned	YES	unset	administratively down	down
FastEthernet2/0	unassigned	YES	unset	administratively down	down

a) Se a mesma rede IP dentro das VLANs, consigo manter conectividade entre estações da mesma VLAN? Se as estações estiverem na mesma VLAN e na mesma rede, sim é possível, pois o tráfego entre a mesma VLAN é encaminhado diretamente pelo trunk.

b) Se a mesma rede IP dentro das VLANs, consigo manter conectividade entre estações de VLANs diferentes?

Não, porque o Router que vai encaminhar o tráfego entre diferentes VLANs, para cada subinterface, só pode ter IPs de redes diferentes configurados em cada subinterface. E sendo que as estações estão em VLANs diferentes, é necessário criar uma nova subinterface numa rede diferente, na qual não é possível configurar o endereço IP na mesma rede. No entanto, se as redes fossem diferentes, seria possível.

